

JINAN, China

WMO: 548230

Lat: 36.6019N

Lon: 117.0081E

Elev: 190

StdP: 14.60

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	20.0	-9.8	3.3	26.5	-6.1	4.0	27.1	19.4	38.6	17.2	37.4	5.2	110	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	95.3	74.0	93.1	74.2	90.9	73.7	80.7	89.6	79.6	88.0	78.4	86.3	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	148.1	85.9	77.2	142.7	85.2	76.0	136.9	84.0	44.7	89.8	43.5	88.1	42.2	86.7	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	16.6	14.3	12.1	99.7	4.0	2.8	9.2	101.7	6.9	103.4	4.6	105.0	1.7	107.0		
			9.2	82.9	3.5	1.4	6.7	83.8	4.6	84.7	2.6	85.4	0.1	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.4	32.2	38.2	49.4	61.6	71.9	79.7	82.0	79.3	72.2	61.9	47.7	35.8
	DBStd	18.75	6.58	7.93	8.81	8.33	6.93	5.63	4.70	4.88	5.70	7.18	8.56	6.81
	HDD50	1607	552	337	118	9	0	0	0	0	0	5	143	443
	HDD65	4022	1017	750	490	160	20	1	0	0	10	145	522	907
	CDD50	5046	0	6	98	358	680	891	992	909	665	373	73	1
	CDD65	1985	0	0	5	58	235	441	527	444	225	48	2	0
	CDH74	19545	0	0	22	385	2051	4946	6079	4330	1533	197	2	0
	CDH80	7911	0	0	2	83	683	2259	2777	1672	411	24	0	0
Wind	WSAvg	6.6	5.9	6.7	7.8	8.4	7.6	7.0	5.9	5.5	5.6	6.1	6.4	5.9
Precipitation	PrecAvg	30.0	0.2	0.6	0.6	1.3	2.8	3.7	8.3	7.3	2.6	1.3	1.0	0.3
	PrecMax	43.7	1.5	1.4	1.7	4.4	6.4	9.1	16.1	16.5	10.6	4.8	4.6	1.3
	PrecMin	17.8	0.0	0.0	0.0	0.3	0.4	0.4	2.0	2.4	0.3	0.0	0.0	0.0
	PrecStd	6.3	0.4	0.4	0.4	0.9	1.5	2.1	3.8	3.2	2.1	1.1	1.1	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	62.7	77.1	87.3	94.8	99.0	98.1	95.6	90.8	83.2	72.6	57.3
		MCWB	39.7	49.8	56.5	65.4	66.0	69.2	75.1	79.0	72.5	63.3	55.1	44.0
	2%	DB	48.9	57.2	71.4	82.0	89.4	95.3	95.2	92.5	87.2	78.9	67.6	52.4
		MCWB	37.7	44.4	53.5	62.3	65.8	69.5	77.4	78.6	70.8	60.9	52.8	41.5
	5%	DB	45.4	53.8	67.4	78.4	86.6	92.9	93.1	90.0	84.5	75.7	63.8	48.9
		MCWB	35.7	42.1	51.3	60.4	64.8	69.5	77.3	77.1	68.8	59.5	50.4	38.9
	10%	DB	42.3	50.6	63.2	74.9	83.8	90.4	90.9	87.9	81.9	72.7	60.3	45.8
		MCWB	33.6	39.8	48.6	58.0	63.6	69.3	76.5	76.2	67.7	58.5	48.7	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	50.7	60.2	69.5	72.0	78.3	82.5	82.3	78.1	67.6	58.3	46.8
		MCDB	49.7	62.4	73.4	81.5	83.1	89.7	92.4	91.1	85.3	75.2	68.4	53.5
	2%	WB	38.7	46.1	55.6	65.3	69.9	75.7	81.0	80.7	74.3	65.3	54.8	43.2
		MCDB	46.8	55.3	68.8	78.4	82.6	85.8	90.3	89.1	82.6	72.9	62.7	49.3
	5%	WB	36.6	43.4	52.6	62.6	68.2	73.9	80.0	79.6	72.0	63.1	52.5	40.8
		MCDB	44.2	52.0	65.3	75.3	80.6	84.9	88.9	87.4	80.7	71.6	61.3	46.8
	10%	WB	34.3	40.6	49.4	59.9	66.3	72.4	78.9	78.4	70.1	60.7	50.0	38.5
		MCDB	41.0	49.5	61.8	72.4	79.6	84.1	87.5	85.3	78.4	69.7	58.7	44.2
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.9	15.5	16.7	16.7	16.5	13.6	13.2	14.9	15.0	13.3	12.6
		MCDBR	16.2	17.0	18.3	18.7	19.0	19.2	16.1	15.3	17.5	17.4	16.1	15.9
		MCWBR	10.2	10.5	9.8	9.1	8.1	6.7	5.7	5.8	7.0	7.5	8.4	9.6
	5% WB	MCDBR	14.4	15.8	17.5	18.6	16.6	15.9	13.6	13.1	14.2	14.3	13.7	13.3
		MCWBR	9.6	10.6	10.4	10.5	8.9	6.9	6.1	5.6	7.1	7.4	9.0	9.3
Clear-Sky Solar Irradiance	taub	0.524	0.597	0.613	0.636	0.660	0.796	0.771	0.738	0.674	0.676	0.589	0.529	
	taud	1.750	1.638	1.631	1.622	1.617	1.431	1.521	1.564	1.632	1.575	1.676	1.763	
	Ebn at Noon	193	195	210	216	214	187	190	192	194	173	172	180	
	Edh at Noon	55	69	76	81	82	99	90	85	75	73	59	51	
All-Sky Solar Radiation	RadAvg	733	926	1353	1604	1803	1667	1477	1408	1261	1016	771	676	
	RadStd	84	138	115	96	130	136	117	155	130	108	117	77	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-0.70	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	N/A

Nomenclature: See separate page